

Jaeryun Joo

AI Engineer

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SUMMARY

AI engineer with two years of backend and infrastructure experience building distributed systems, APIs, and data processing workflows. Experience includes three AI projects involving LLM evaluation, prompt refinement, RAG corpus design, FastAPI-based AI features, and cloud deployment. **Focused on connecting LLMs with backend services and data pipelines to build reliable, evidence-grounded AI applications.**

SKILLS

- **AI & LLM:** Azure OpenAI, RAG, Prompt Engineering, LLM Evaluation, Azure AI Search, Vector Search
- **Backend & Web:** Python, FastAPI, TypeScript, Next.js, React, NestJS, Express, REST APIs
- **Data:** SQL, PySpark, Pandas, NumPy, Databricks, Azure Data Factory, ADLS Gen2, Microsoft Fabric, Event Hubs
- **Databases:** PostgreSQL, Microsoft SQL Server, MongoDB, Redis, LevelDB
- **Infrastructure & Operations:** Linux, Azure, AWS EC2, Docker, Bicep, GitHub Actions, Prometheus, Grafana
- **Additional:** Go, JavaScript, PHP, C/C++, Git, GitHub, Jira, Swagger, Postman

EXPERIENCE

EQBR Holdings | Blockchain Research Engineer | Jan 2021 - Dec 2022

- Migrated a legacy PHP blockchain client to a Go-based open-source client and extended consensus-related data and messaging structures, increasing throughput from **3,000 to 4,500 TPS**
- Separated query workloads from consensus nodes by building a ledger replica and a CQRS-based relational serving layer
- Designed and implemented an event-sourcing mechanism that reconstructs application state at a specified block to resolve state inconsistencies between the backend and blockchain
- Automated multi-node P2P test environment provisioning with Docker Compose, reducing setup time by **more than 50%**
- Built node-state visualization and Grafana health checks for real-time monitoring of consensus progress, P2P latency, and database usage

PROJECTS

Public Procurement Bid Requirement Analysis & Response Automation | Data & LLM Engineer | MS Data School | May 2026

- Built a **five-stage AI service pipeline** covering procurement notice and attachment ingestion, LLM-based requirement extraction, RAG-based question answering, and company-specific opportunity matching
- Evaluated gpt-4.1-nano and gpt-4.1-mini on the same real-world notices; nano extracted no conditions from two documents, while mini extracted **7 and 19**, respectively, leading to a model switch after weighing missed requirements against the added cost
- Refined prompts with explicit decision criteria and positive and negative examples, then added regex and keyword validation to reduce misclassification of similar bid conditions
- Standardized the deployment of **eight Azure resources** with Bicep and GitHub Actions; integrated Application Insights and Microsoft Teams to capture execution logs and deliver failure alerts

Post-Offboarding Residual Access Detection & Remediation Support | Data Engineer | MS Data School | Jun - Jul 2026

- Integrated **16 data objects across seven IdP, IAM, and SaaS systems** into a unified access graph and designed a RAG data layer that supplies supporting evidence for remediating residual access
- Normalized policies, remediation procedures, and official technical guides into a shared RAG schema; implemented a data contract and ingestion pipeline with source attribution and change tracking
- Excluded raw credential values and duplicate records before ingestion to prevent sensitive data from entering the RAG corpus or LLM context
- Designed **seven security remediation scenarios for RAG evaluation** and defined the expected supporting evidence for each

Vulnerability Reassessment & Remediation Automation | Data & Full-Stack Engineer | MS Data School | Aug - Sep 2026

- Built a FastAPI endpoint that retrieves only the operational data relevant to each query and organization, constructs the LLM context, and returns evidence-grounded remediation guidance
- Normalized records and analysis documents from multiple tables into a common grounding format and returned only the evidence identifiers and source links cited in each answer
- Revalidated the organization scope of every grounding source immediately before the LLM call, blocking response generation when data from another organization was detected
- Consolidated **406 vulnerability alerts into 10 package-level review and remediation targets**

EDUCATION

Sungkyunkwan University | B.S. in Software Engineering | Expected Feb 2027

- GPA: 3.79/4.5 · Dean's List, Fall 2020 (GPA 4.5/4.5)

CERTIFICATIONS & ACTIVITIES

Microsoft Certified: Azure Data Fundamentals (DP-900) | Aug 2026

MS Data School - Data Processing & Analytics Program | Mar - Sep 2026

- Led three team projects; placed **1st among five teams** in the first project
- Organized a four-person Python and SQL study group

ADDITIONAL INFORMATION

- **Military Service:** Completed mandatory service as a Sergeant in the Republic of Korea Army (Information and Communications) | Sep 2023 - Mar 2025
- **English:** OPIc IM2 | Sep 2026
- **Open Source:** Submitted a pull request improving code comments in Ethereum Geth | Jun 2021
- **Industry-Academic Collaboration:** EDA and preprocessing of system logs for failure prediction, Sungkyunkwan University–Shinhan Bank | Apr - Nov 2019